



David Pascual-Hernández

COMPUTER VISION ENGINEER & RESEARCHER

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Summary

Computer Vision Engineer and Researcher with 6+ years of experience designing, developing, and deploying end to end machine learning and computer vision systems. Experienced across the full ML lifecycle, from data acquisition and dataset curation to model development, evaluation, optimization, and deployment under real world constraints. Previously led the R&D Optical Team at SEDDI while remaining hands on in software engineering and model development. Currently pursuing a PhD at Rey Juan Carlos University focused on multimodal perception using RGB, LiDAR, and sensor fusion for autonomous navigation in off-road environments. Strong background spanning classical computer vision, deep learning, and applied research with a track record of shipping production systems.

Work Experience

Rey Juan Carlos University

Madrid, Spain

RESEARCH ASSISTANT

Oct. 2024 – Present

- Research on deep learning models for real time **semantic segmentation** using **RGB, LiDAR**, and **multimodal sensor fusion** for robust perception and **autonomous navigation** in challenging off road environments, with a focus on domain adaptation and cross-dataset evaluation.
- Developed synthetic data generation pipelines in **CARLA** to create diverse virtual driving scenarios for training and evaluating perception models under controlled environmental conditions.

SEDDI

Madrid, Spain

OPTICAL TEAM LEAD

Jan. 2023 – Oct. 2024

- **Led an R&D team** of computer vision engineers while remaining hands on across the full machine learning lifecycle, from product requirements and dataset acquisition to model development, deployment, and production validation.
- Developed and deployed deep learning models based on **Transformers** and **GANs** for image restoration and image to image translation, balancing accuracy, inference latency, robustness, and scalability.
- Worked closely with product management to translate business requirements into production ready computer vision solutions, driving experimentation, evaluation, and deployment decisions.

SEDDI

Madrid, Spain

COMPUTER VISION ENGINEER

Aug. 2018 – Jan. 2023

- Contributed to the architecture, development, and scaling of **textura.ai** from its inception to early production deployments.
- Designed components of a **multi camera**, multi illumination **gonioreflectometer** for production grade textile material digitization.
- Developed robust **color constancy** and calibration pipelines for imaging systems operating under strict reliability and performance constraints.

Nokia Bell Labs

Madrid, Spain

IMMERSIVE VIDEO RESEARCHER INTERN

Oct. 2017 – Aug. 2018

- Developed hand gesture recognition interfaces for human computer interaction.
- Researched augmented and mixed reality technologies for immersive video applications.

Robert Bosch

Madrid, Spain

MACHINE VISION ENGINEER INTERN

Aug. 2016 – Jan. 2017

- Evaluated industrial Data Matrix reading systems, imaging hardware, and optical lenses.
- Designed and validated machine vision inspection systems using NeuroCheck and custom optical design tools.

Education

Rey Juan Carlos University

Madrid, Spain

PHD IN ARTIFICIAL VISION, PATTERN RECOGNITION AND IMAGE PROCESSING

2024 – Present

- Research focused on multimodal perception, semantic segmentation, and robust autonomous navigation in unstructured environments.

Rey Juan Carlos University

Madrid, Spain

MSc IN COMPUTER VISION

2017 – 2020

- Master's Thesis: **Efficient 3D Human Pose Estimation from RGBD Sensors**

Rey Juan Carlos University

Madrid, Spain

BSc IN AUDIOVISUAL AND MULTIMEDIA SYSTEMS ENGINEERING

2012 – 2017

- Bachelor's Thesis: **Study of Convolutional Neural Networks using Keras**

Óbuda University

Budapest, Hungary

ERASMUS+ IN ELECTRICAL ENGINEERING

2016

Certifications & Continuous Education

2026	Summer School , iRoboCity2030: ROS 2, AI and Field Robotics	Madrid, Spain
2025	Summer School , ESSAI: European Summer School on Artificial Intelligence	Bratislava, Slovakia
2025	Certification , Introduction to Self-Driving Cars (Coursera)	Online
2024	Certification , Cambridge Linguaskill General (CEFR C1 Level)	Madrid, Spain
2021	Certification , Full Stack Deep Learning (The Full Stack)	Online

COMPETITIONS

2026	5th Place , GOOSE 2D Fine-Grained Semantic Segmentation Challenge	Workshop on Field Robotics, IEEE ICRA
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Skills

ML & CV	PyTorch, TensorFlow, OpenCV, ONNX, CNNs, Transformers, LiDAR, Sensor Fusion, Semantic Segmentation
Deployment	Docker, ONNX Runtime, Weights & Biases, GitLab/GitHub CI/CD, Streamlit
Programming	Python (including relevant scientific libraries), ROS 2, C++ (working knowledge), MATLAB, LaTeX
Tools	Linux, Git, VS Code, Elasticsearch, Kibana

Publications

Cross-Dataset Evaluation of Visual Semantic Segmentation Models for Off-Road Autonomous Driving 2026

EXPERT SYSTEMS WITH APPLICATIONS

David Pascual-Hernández, Sergio Paniego, Roberto Calvo-Palomino, Ignacio Mora-Jiménez, José María Cañas-Plaza

Deep Learning-Based Semantic Segmentation of LiDAR Point Clouds for Autonomous Driving in Unstructured Environments 2026

INTERNATIONAL WORKSHOP ON PHYSICAL AGENTS (WAF)

Félix Martínez-Ortega, David Pascual-Hernández, Daniel Borja-Fernández, Inmaculada Mora-Jiménez, José María Cañas-Plaza

Single-image Reflectance and Transmittance Estimation from Any Flatbed Scanner 2025

COMPUTERS & GRAPHICS

Carlos Rodríguez-Pardo, David Pascual-Hernández, Javier Rodríguez-Vázquez, Jorge López-Moreno, Elena Garcés

Towards Material Digitization with a Dual-scale Optical System 2023

ACM TRANSACTIONS ON GRAPHICS (SIGGRAPH)

Elena Garcés, Víctor Arellano, Carlos Rodríguez-Pardo, David Pascual-Hernández, Sergio Suja, Jorge López-Moreno

UMat: Uncertainty-Aware Single Image High Resolution Material Capture 2023

IEEE/CVF CONFERENCE ON COMPUTER VISION AND PATTERN RECOGNITION (CVPR)

Carlos Rodríguez-Pardo, Henar Domínguez-Elvira, David Pascual-Hernández, Elena Garcés

Efficient 3D Human Pose Estimation from RGBD Sensors 2022

DISPLAYS

David Pascual-Hernández, Noelia O. De Frutos, Ignacio Mora-Jiménez, José María Cañas-Plaza

Depth from Focus: An Application for Fabrics Captured at Microscale 2019

CVPR WORKSHOPS

David Pascual-Hernández, Elena Garcés, Carlos Aliaga, Jorge López-Moreno

Automatic Extraction and Synthesis of Regular Repeatable Patterns 2019

COMPUTERS & GRAPHICS

Carlos Rodríguez-Pardo, Sergio Suja, David Pascual-Hernández, Jorge López-Moreno, Elena Garcés

Extracurricular Activities

SEO/BirdLife

VOLUNTEER IN CITIZEN SCIENCE INITIATIVES Nov. 2023 – Present

Google Summer of Code

MENTOR & ORG ADMIN FOR JdeROBOT Jun. 2020 – Present

- Provide technical leadership and **mentorship** to international students in **open-source software development** frameworks.
- Guide high-level project architecture, conduct rigorous **code reviews**, and oversee technical implementation and community collaboration.

JdeRobot

OPEN SOURCE CONTRIBUTOR Sep. 2017 – Present

- Develop open-source educational and research tools related to **robotics**, **computer vision**, and **autonomous systems**.
- Design modular, reusable infrastructure components to support developer ecosystem scaling.